

M5 Volume-Confirmed Reversal: Research Report

RegimeForgeEA

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Executive summary

This report documents a pre-specified M5 long reversal candidate that passed 2021-2023 training, separate 2024 validation, and untouched 2025 holdout testing on public PAXGUSDT data. PAXGUSDT is a gold-linked proxy, not broker-native XAUUSD. This is therefore a research candidate, not a claim of XAUUSD or live profitability.

Trading rule

Let C_t and V_t be M5 close and volume. Define:

$$r_3(t) = \frac{C_t}{C_{t-3}} - 1$$

$$q_{20}(t) = Q_{0.20}(\{r_3(s) : s < t\})$$

The quantile uses the preceding 20 trading days (5,760 M5 bars), shifted one bar so it contains no current-bar information. Define:

$$VR(t) = \frac{V_t}{SMA_{60}(V)_t}$$

Open a long position only when:

$$r_3(t) \leq q_{20}(t) \quad \wedge \quad VR(t) \geq 1.5$$

Evaluate at M5 close, enter at the next M5 open plus half of a fixed 0.35 USD round-trip spread, exit at the close of bar $t + 24$ minus half spread, and prohibit position overlap. The holding period is 120 minutes.

Research protocol

The rule was selected only from 2021-2023 after a fixed direct order-level grid covering order-flow, return-reversal, breakout-reversal, and volume-confirmed reversal families; both directions; four UTC sessions; and 5-120 minute exits. Training gates: 100+ trades, PF ≥ 1.10 , and mean net return ≥ 0.50 bps. Validation gates: 30+ trades, PF ≥ 1.05 , and mean net return ≥ 0.25 bps. The 2025 holdout was inspected only after 2024 validation passed.

Performance results

PnL uses 0.10 lot and 100 oz only to express dollars. PF, win rate, and bps are position-size independent under the fixed-cost model.

Sample	Trades	Net PnL (USD)	PF	Win rate	Mean net bps	Max drawdown (USD)
Training (2021-2023)	4,333	11,644.50	1.156	51.63%	1.512	-2,111.00
Validation (2024)	1,428	7,832.00	1.257	55.04%	2.447	-1,847.00
Holdout (2025)	1,350	6,707.20	1.119	55.41%	1.768	-4,927.40

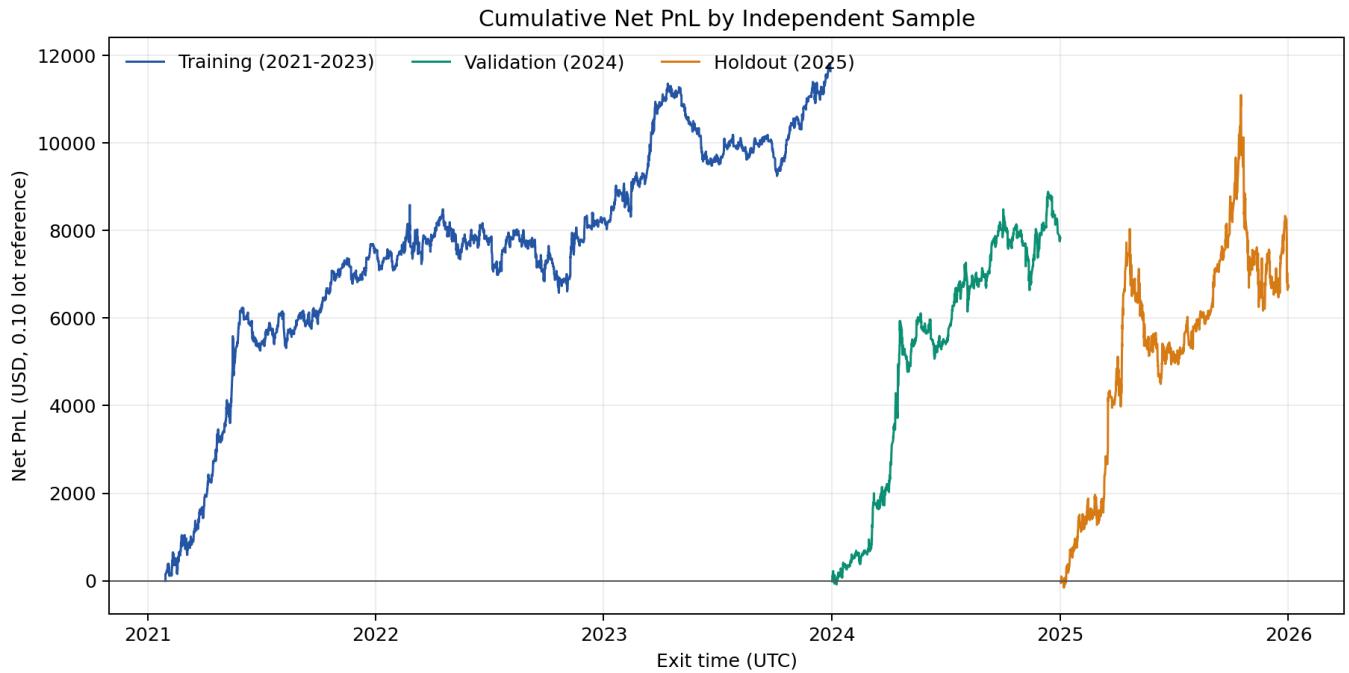


Figure 1: Cumulative PnL by independent sample

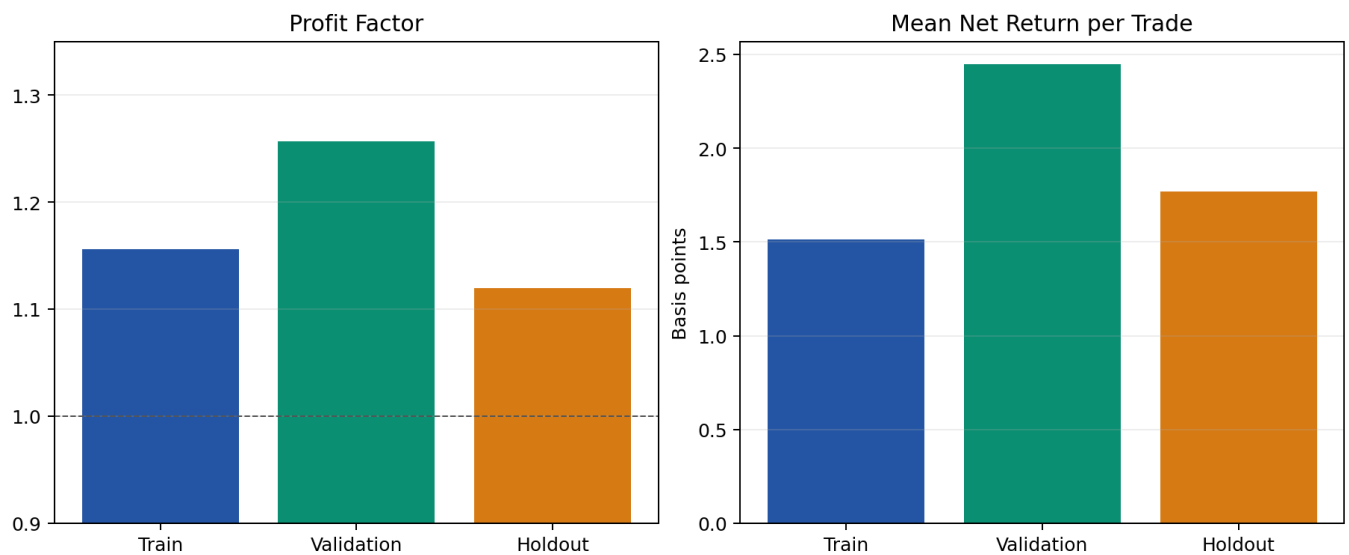


Figure 2: Profit factor and mean return

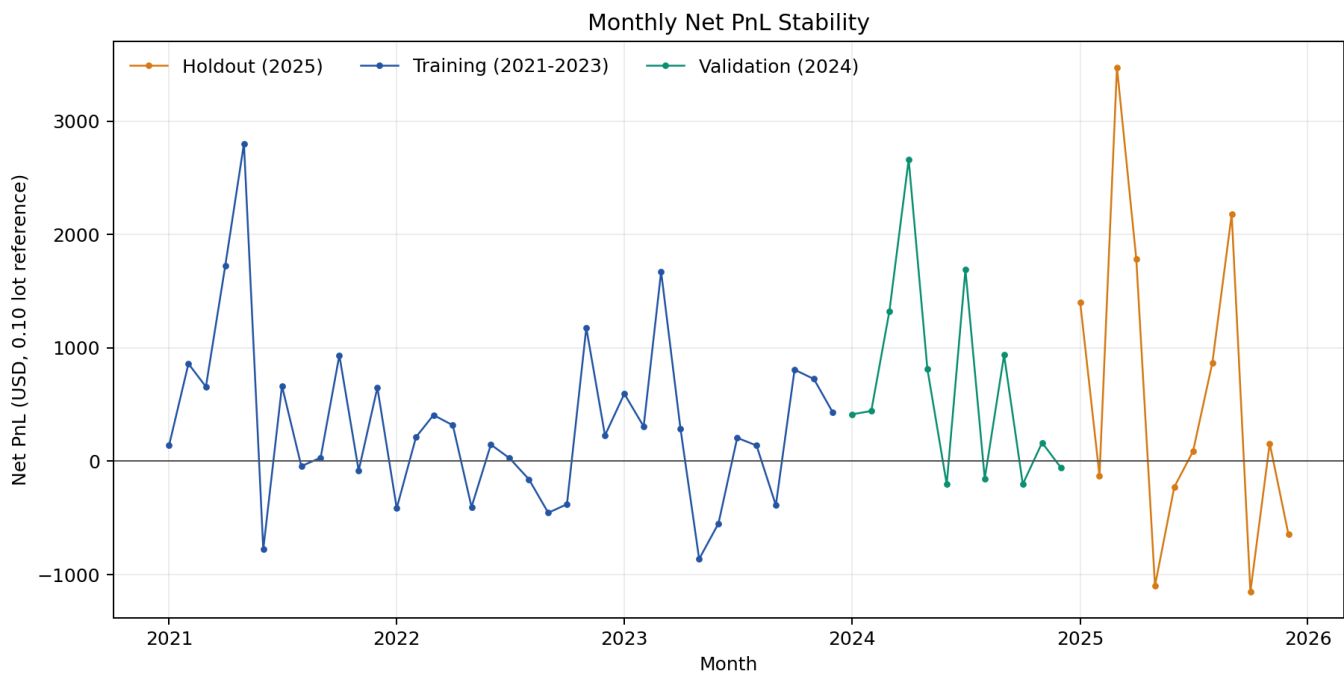


Figure 3: Monthly PnL stability

Interpretation and professional-test handoff

All three independent proxy samples are positive after the fixed-spread assumption: training PF 1.156, validation PF 1.257, and holdout PF 1.119. The 2025 result is weaker than validation and supports conservative risk gates.

This work does **not** test broker-native XAUUSD Bid/Ask bars, real fill latency, slippage, commissions, swaps, symbol-specific contract parameters, weekend gaps, or order rejection. A professional tester should reproduce the exact rule on broker M5 or tick Bid/Ask history before demo or live use. The EA must remain disabled for new entries until that test passes.

Reproducibility

```
python scripts/research_m5_order_flow_grid.py \
  data/derived/PAXGUSDT_5m_2021_2025_weekdays.csv \
  data/derived/PAXGUSDT_order_flow_5m_2021_2025.csv \
  --output outputs/m5_executable_factor_grid.csv \
  --report reports/M5_Executable_Factor_Grid.md

python scripts/research_order_flow_absorption.py \
  data/derived/PAXGUSDT_5m_2021_2025_weekdays.csv \
  data/derived/PAXGUSDT_order_flow_5m_2021_2025.csv \
  --factor volume_return_3_reversal --bar-minutes 5 --hold-bars 24 \
  --side long --session-hours 0-23 \
  --trades outputs/m5_volume_reversal_holdout_trades.csv \
  --report reports/M5_Volume_Reversal_Holdout.md
```